Universitat Oberta de Catalunya

UOC's Journey with OpenAlex:

Data analysis, workflows and transparency from an institutional perspective

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OpenAlex Users Meeting,
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uoc.edu

Who We Are & Why We Work with OpenAlex

1. Who We Are: Background & Context

Open Science Office (UOC) working with OpenAlex

- **2022–2023** Initial exploration & first report
- **2024** Exploratory Power BI dashboard
- **2025** Repository audit, CWTS stay & Leiden Ranking
- **2026** Advanced automation & audits of profiles and coverage

Outreach & Collaboration

Community Outreach: Collaborative initiatives (Spanish University Libraries Network REBIUN, Leiden Ranking) & staff training.
Building Expertise: Extraction & analysis tools (API, Python, RStudio).

2. Why OpenAlex? & Objectives

Commitment to Open Science

Democratization
Open data, free for any person or institution, with no costly subscriptions.

Transparency
Open, reproducible audits and studies of science.

Inclusion
Wider coverage of Global South research and non-English languages.

Key Objectives

1. **Workflow Integration:** Evaluate OpenAlex in UOC workflows.
2. **Metadata Analysis:** Coverage analysis & metadata error detection.
3. **Visibility Improvement:** O2 repository optimization & monitoring.

Auditing our institutional repository (O2) in OpenAlex

1 Objectives & Motivation

- Ensure open research **visibility**
- Strategic alignment (**CoARA & Leiden Ranking**)
- Detect **metadata loss**

2 Methodology & Tools

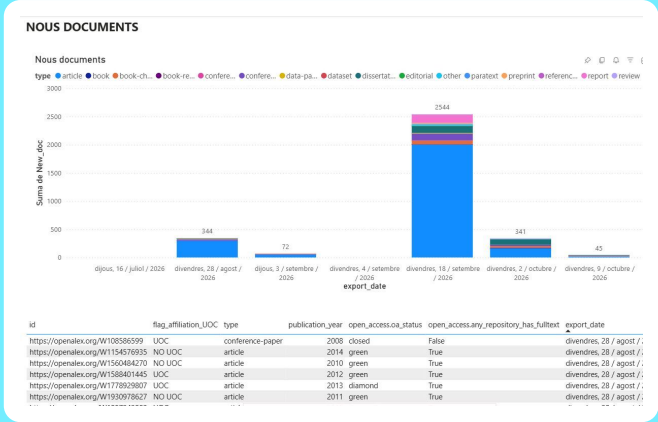
- Automated **extraction** & **dashboards**
- Mass queries in SQL environment (CWTS Stay 2025)
- Periodic **metadata audits**

3 Results

- Abrupt **drop in Handles** (Decay)
- The **Shadowed Green phenomenon**
- **Inconsistencies** in the Source entity

4 Conclusions & Actions

- **Curation** at source **(Pre-OpenAlex)**
- Direct **indexing** requests
- Essential **"Post-OpenAlex"** filtering
- **Open** tools & **collaborative** network



id	flag_affiliation_UOC	type	publication_year	open_access.oa_status	open_access.any_repository_has_fulltext	export_date
https://openalex.org/W108586599	UOC	conference-paper	2008	closed	False	divendres, 28 / agost /
https://openalex.org/W1154576935	NO UOC	article	2014	green	True	divendres, 28 / agost /
https://openalex.org/W1560484270	NO UOC	article	2010	green	True	divendres, 28 / agost /
https://openalex.org/W1588401445	UOC	article	2012	green	True	divendres, 28 / agost /
https://openalex.org/W1778929807	UOC	article	2013	diamond	True	divendres, 28 / agost /
https://openalex.org/W1930978627	NO UOC	article	2011	green	True	divendres, 28 / agost /

Number of new documents added from OpenAlex to O2, the institutional repository of the Universitat Oberta de Catalunya (UOC), per export date and broken down by document type (July–October 2026).

UOC's Positioning in the CWTS Leiden Ranking Open Edition

1 Objectives & Motivation

- Historical **transition** of **CWTS** towards **Open Science**
- Alignment with the **Barcelona Declaration** and **CoARA**
- Assess the actual **visibility** of UOC

2 Methodology & Tools

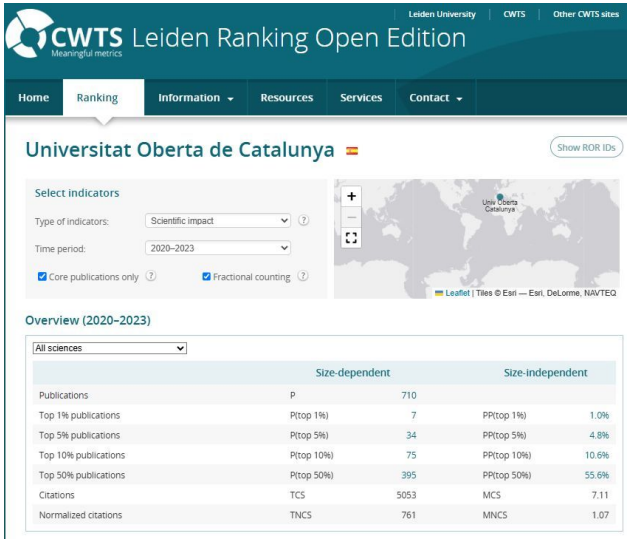
- Analysis of the **2020-2023 period** (2024-2025 Editions)
- **Cross-referencing DOIs** with corporate CRIS
- SQL queries directly on CWTS's database (2025 Stay)

3 Results

- **Global volume** very close to **CRIS**
- Leadership in **Gold Open Access** (45%)
- High **global Open Access** (72.1%)
- **Identification** of 355 "lost" publications

4 Conclusions & Actions

- Immediate **recovery of 219 publications**
- Opportunities with the **"Non-Core" version** (2025)
- Curation at source (**Pre-OpenAlex**)



UOC scientific impact indicators in the CWTS Leiden Ranking Open Edition 2025.

Auditing UOC's Author Profiles and Affiliations

1 Objectives & Motivation

- Identify misaffiliated authors and works linked to UOC
- **Detect profiles requiring review** or merging
- Assess completeness and **reliability of author metadata**

2 Methodology & Data

- **Reproducible R pipeline** (openalexR) fetching UOC-affiliated works (dissertations excluded)
- Cross-referenced with CRIS Data **via ORCIDs**
- Fully reusable framework to refresh results periodically

3 Results

~3% of profiles list 8+ institutions

~3% duplicate ORCIDs and **~2%** duplicate normalized names

65% of UOC-affiliated ORCIDs in OpenAlex are not in our CRIS; they account for 31% of those authors' publications (to be reviewed)

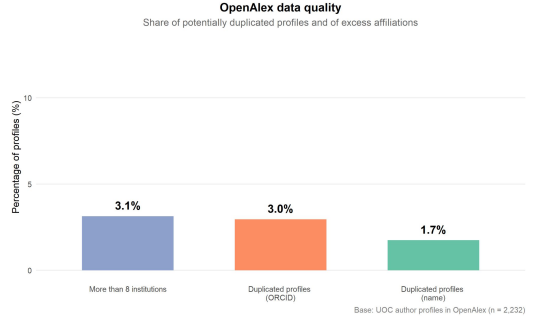
4 Conclusions & Next Steps

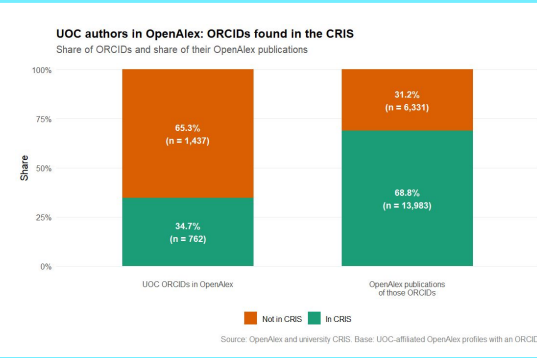
Positive trend from regular audits, but data cleanup is still required.

Next steps:

- Pilot with researchers to clean "noisy" profiles
- Join OpenAlex as a member to access better curation tools

Full report
Last updated: 16 October 2026





CRIS vs. OpenAlex: Coverage & Metadata Consistency
DOI-based comparison of UOC (2010-2025)

1 Objectives & Motivation

- Coverage analysis
- **Explain differences in coverage** and detect possible biases
- **Test metadata** on matched publications

2 Methodology & Data

- **Reproducible R Markdown report;** works from OpenAlex via openalexR (dissertations excluded)
- Cross-referenced with CRIS / Data Warehouse **via DOIs**
- Metadata checks on 5,881 shared DOIs

3 Results

Coverage: 51% of OpenAlex works are also in our CRIS (art 55%, data 4%)

Gaps:

- Misaffiliations
- Noisy records for datasets and non-traditional work types
- Outreach journals

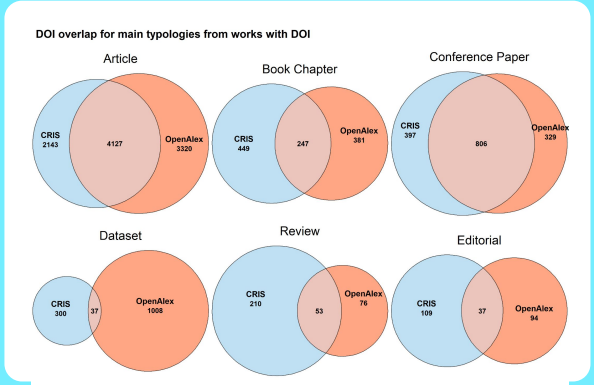
OA Bias: OpenAlex-only works are 77% OA vs 57% for matched works

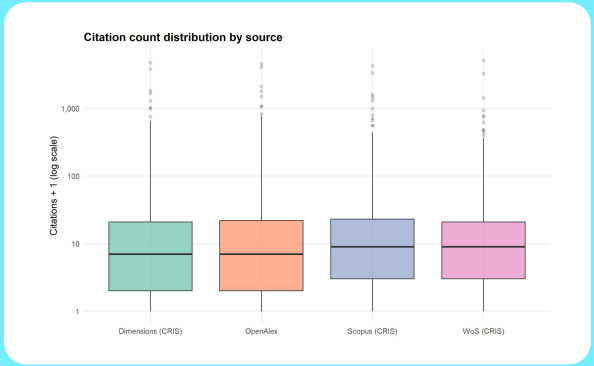
Metadata: High agreement in Year (98%), Lang (95%), Type (91%), OA (78%)

4 Conclusions & Next Steps

- Metadata consistency is overall high
- Datasets, outreach journals, and misaffiliations drive OA overestimation
- Need to curate works and author affiliations on OpenAlex which would substantially improve results

Full report
Last updated: 16 October 2026





General Conclusions

1 Main Conclusions

Curation

- Community & curation at source
- Human curation layer to improve trust

Improvements

- Focus on profile & affiliation disambiguation
- Noisy record indexing of datasets and other minority work types.

Analysis

- Users must apply their own filters and indicators

2 Wishlist

Broader Curation

- More accessible Work-level curation tools

Comprehensibility

- Transparent, documented backend processes

More data labels

- Labels such as peer-reviewed vs. opinion content

3 Next Steps

Strategic Integration

- Embed OpenAlex data in our CRIS and repository workflows

Action Plan

- Implement curation processes
- Continuous monitoring & learning
- Share results across networks

Universitat Oberta de Catalunya

Q&A

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